

2009 F=ma Exam: Problem 7

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We use the kinematics equation without the time variable,

$$v_f^2 = v_0^2 + 2ax$$

The acceleration is

$$a = \frac{v_f^2 - v_0^2}{2x} = \frac{(15 \text{ m/s})^2 - (10 \text{ m/s})^2}{2(25 \text{ m})} = 2.5 \text{ m/s}^2$$

so the answer is B.